

Homework assignment 3

Due Date: 29th of November EOD

Course: Stochastic Methods in Mathematical Modelling and AI

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Problem 1. Simulation of Probability Flow ODE (4 points)

In this task we will consider a simple 1D data distribution which is a mixture of two Gaussians with variance s^2 and located at points m and $-m$ (take $s = 1$ and $m = 5$):

$$p_{\text{data}}(x) = \frac{1}{2}\mathcal{N}(x|-m, s^2) + \frac{1}{2}\mathcal{N}(x|m, s^2)$$

For a samples from $p_{\text{data}}(x)$ we will apply the Itô process:

$$dX_t = \mu(X_t, t) dt + \sigma(t) dW_t$$

with $\mu(X_t, t) = -X_t$ and $\sigma(t) = \sqrt{2}$.

1. Write a numerical simulation of this Itô process (forward SDE). For discretization use $dt = h$, $dW_t = \sqrt{h} \cdot \xi$, where $\xi \sim \mathcal{N}(0, 1)$ for some small h . Sample 100 points from $p_{\text{data}}(x)$ and plot for each of them the SDE trajectory on the same plot. Make sure that the process converged to a normal distribution by plotting histogram of the final positions. **(1 point)**
2. Find analytical solution $p(x, t)$ of the corresponding Fokker-Planck equation, and plot $p(x, t)$ for the same range as your numerical simulation. Hint: you can use the solution of the heat (diffusion) equation and apply change of variables from the lecture 3 on diffusion models. **(1 point)**
3. Using the analytical solution, compute $\frac{d}{dx} \log p(x, t)$ (numerically using finite differences) and numerically simulate the probability flow ODE (forward in time):

$$dX_t = \left[\mu(X_t) - \frac{\sigma(t)^2}{2} \frac{d}{dx} \log p(X_t, t) \right] dt,$$

where $\mu(X_t, t) = -X_t$ and $\sigma(t) = \sqrt{2}$. For the same 100 points from $p_{\text{data}}(x)$ as in the first step, plot the corresponding probability flow ODE trajectories for each point. **(2 points)**

Problem 2. Lipschitz constant bounds for the score function. (4 points)

In this task we will study behaviour of score function $\nabla \log p(x, t)$. By definition, Lipschitz constant of a function equals to the maximal absolute value of its gradient. So, for the score function we get

$$\begin{aligned} L &= \sup_{x \in \mathbb{R}^d} \|\nabla(\nabla \log p(x, t))\| = \sup_{x \in \mathbb{R}^d} \left\| \nabla \frac{\nabla p(x, t)}{p(x, t)} \right\| = \sup_{x \in \mathbb{R}^d} \left\| \frac{\nabla^2 p(x, t)}{p(x, t)} - \frac{(\nabla p(x, t))(\nabla p(x, t))^{\top}}{p(x, t)^2} \right\| \leq \\ &\leq \sup_{x \in \mathbb{R}^d} \left(\frac{\|\nabla^2 p(x, t)\|}{p(x, t)} + \frac{\|\nabla p(x, t)\|^2}{p(x, t)^2} \right) \text{ by triangle inequality.} \end{aligned}$$

We already established bounds for $\|\nabla p(x, t)\|$, so in order to bound the score function we need to study behaviour of the second derivative $\|\nabla^2 p(x, t)\|$ (i.e. the Hessian of $p(x, t)$).

Lets start from the heat equation. Consider the heat kernel

$$G_t(x) = \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{|x|^2}{4t}\right), \quad t > 0, x \in \mathbb{R}^d.$$

1. Show that second derivatives of the heat kernel have the form (1 point)

$$\nabla_x^2 G_t(x) = \left(\frac{xx^\top}{4t^2} - \frac{1}{2t} I_d \right) G_t(x).$$

2. Using formula for second derivatives and a method from the lecture 2 on diffusion models show that there exists a constant $C_d^{(2,\infty)} > 0$ such that for every $t > 0$, (1 point)

$$\sup_{x \in \mathbb{R}^d} |\nabla_x^2 G_t(x)| \leq C_d^{(2,\infty)} t^{-\frac{d+2}{2}}. \quad (1)$$

3. Let $u_0 \in L^1(\mathbb{R}^d)$ and let u solve the heat equation with initial data u_0 . Show using the previous bounds and a method from the lecture 2 on diffusion models that for every $t > 0$,

$$\|\nabla_x^2 u(t, \cdot)\|_{L^\infty(\mathbb{R}^d)} \leq C_d^{(2,\infty)} t^{-\frac{d+2}{2}} \|u_0\|_{L^1(\mathbb{R}^d)} \quad (2)$$

Why the score function will have a high Lipschitz constant if the density $p(x, t)$ is low? (2 points)

Problem 3 Optimal transport with rectified flows (4 points)

The rectified flow aims to approximately solve the problem of **optimal transport** from the initial distribution π_0 to the target distribution π_1 . It means that we want to find a **transport plan** in the form of a joint distribution $\pi(X_0, X_1)$ which minimizes the quadratic transport cost

$$\mathbb{E}(X_1 - X_0)^2 = \int \pi(X_0, X_1)(X_1 - X_0)^2 dX_0 dX_1,$$

also for a transport plan we require $\int \pi(X_0, X_1) dX_0 = \pi_1(X_1)$ and $\int \pi(X_0, X_1) dX_1 = \pi_0(X_0)$. For a deterministic **transport map** $X_1 = T(X_0)$ from π_0 to π_1 the transport plan becomes

$$\pi^T(X_0, X_1) = \pi_0(X_0) \delta(X_1 - T(X_0)).$$

To gain a better understanding of the problem formulation of optimal transport through some simple examples, you can read the blog post at <https://alexhwilliams.info/itsneuronalblog/2020/10/09/optimal-transport/>.

Now, let's return to the topic of rectified flows. In the rectified flow approach, the first step involves sampling independent pairs (X_0, X_1) from the most naive transport plan one could ever imagine $\pi_{\text{naive}}(X_0, X_1) = \pi_0(x) \times \pi_1(x)$. The idea is that we completely ignore any knowledge of distances and to transport some point X_0 from the initial distribution π_0 we just choose some random point X_1 from the target distribution π_1 . For example, if you are hungry and your task is to transport yourself from your home to a grocery store, you would pick a random grocery store from anywhere on Earth (*there are grocery stores in Antarctica*) and just go there to buy some food. Of course, this naive transport plan is nowhere to be optimal. We then define the linear interpolation $X_t = tX_1 + (1-t)X_0$ between the initial and target samples. Subsequently, we train a neural network $v_\theta(X_t, t)$ to minimize the loss

$$\min_{\theta} \int_{t=0}^1 \mathbb{E}_{(X_0, X_1) \sim \pi_{\text{naive}}(X_0, X_1)} \left[\|X_1 - X_0 - v_\theta(X_t, t)\|^2 \right] dt.$$

Given a trained rectified flow, we can transport samples from the initial distribution π_0 to the target distribution π_1 in deterministic way by numerically solving the ordinary differential equation (ODE)

$$\frac{dZ_t}{dt} = v_\theta(Z_t, t)$$

for $t \in [0, 1]$ with $Z_0 \sim \pi_0$. In other words, for this particular case the deterministic transport map becomes

$$T_{1\text{-rectified}}(Z_0) = Z_0 + \int_{t=0}^1 v_\theta(Z_t, t) dt.$$

We know, that deterministic transport map $T_{1\text{-rectified}}$ gives rise to the deterministic transport plan $\pi_{1\text{-rectified}}$ which has the explicit form

$$\pi_{1\text{-rectified}}(X_0, X_1) = \pi_0(X_0) \times \delta(X_1 - T_{1\text{-rectified}}(X_0)).$$

It turns out that this deterministic transport plan has much lower transport cost than our extremely naive transport plan $\pi_{\text{naive}}(X_0, X_1)$ (*and you will show it in the task 2 of this assignment*). If you are still reading this assignment you deserve a *hint for the task 2*: the loss for $v_\theta(X_t, t)$ is suspiciously similar to the direct optimization of the transport cost

$$\begin{aligned} \min_{\theta} \mathbb{E}(X_1 - X_0)^2 &= \min_{\theta} \mathbb{E}_{(X_0, X_1) \sim \pi_{\text{naive}}(X_0, X_1)} [\|X_1 - T_{1\text{-rectified}}(X_0)\|^2] = \\ &= \min_{\theta} \mathbb{E}_{(X_0, X_1) \sim \pi_{\text{naive}}(X_0, X_1)} \left[\left\| X_1 - X_0 - \int_{t=0}^1 v_\theta(X_t, t) dt \right\|^2 \right] \end{aligned}$$

In this assignment, you will demonstrate how the rectified flow beginning with the naive transport plan $\pi_{\text{naive}}(X_0, X_1) = \pi_0(x) \times \pi_1(x)$ utilizes this plan to create a more effective deterministic transport plan, denoted as $\pi_{1\text{-rectified}}(X_0, X_1) = \pi_0(X_0) \times \delta(X_1 - T_{1\text{-rectified}}(X_0))$. Subsequently, this 1-rectified flow is used to derive another slightly improved transport plan, $\pi_{2\text{-rectified}} = \pi_0(X_0) \times \delta(X_1 - T_{2\text{-rectified}}(X_0))$, which incurs a have transport cost than the previous plan $\pi_{2\text{-rectified}}$, and so on. This was the general introduction to why rectified flows are useful. Now, to the concrete tasks:

1. Prove that if $X_0 \sim \pi_0(X_0)$, then for any deterministic transport map $X_1 = T(X_0)$, such that $T(X_0) \sim \pi_1$, the corresponding joint distribution $\pi^T(X_0, X_1) = \pi_0(X_0) \times \delta(X_1 - T(X_0))$ satisfies the conditions

$$\int \pi^T(X_0, X_1) dX_0 = \pi_1(X_1)$$

and

$$\int \pi^T(X_0, X_1) dX_1 = \pi_0(X_0),$$

which means that $\pi^T(X_0, X_1)$ is indeed a transport plan. **(1 point)**

2. Prove that the 1-rectified flow indeed reduces the transport cost: for $Z_0 \sim \pi_0$ and $Z_1 = Z_0 + \int_{t=0}^1 v_\theta(Z_t, t) dt$, the quadratic transport cost for $(Z_0, Z_1) \sim \pi_{1\text{-rectified}}$

$$\mathbb{E}(Z_1 - Z_0)^2 \leq \mathbb{E}(X_1 - X_0)^2$$

will not be greater than the transport cost for the initial naive samples $(X_0, X_1) \sim \pi_{\text{naive}}$. The explicit formulas for the transport costs are

$$\begin{aligned} \mathbb{E}(Z_1 - Z_0)^2 &= \int \pi_0(Z_0) \delta \left(Z_1 - Z_0 - \int_{t=0}^1 v_\theta(Z_t, t) dt \right) (Z_1 - Z_0)^2 dZ_0 dZ_1 = \\ &= \int \pi_0(Z_0) \delta(Z_1 - T_{1\text{-rectified}}(Z_0)) (Z_1 - Z_0)^2 dZ_0 dZ_1 = \int \pi_{1\text{-rectified}}(Z_0, Z_1) (Z_1 - Z_0)^2 dZ_0 dZ_1 \end{aligned}$$

where the delta-function has the property $\int \delta(Z - a)f(Z)dZ = f(a)$ and

$$\begin{aligned}\mathbb{E}(X_1 - X_0)^2 &= \int \pi_0(X_0)\pi_1(X_1)(X_1 - X_0)^2 dX_0 dX_1 = \\ &= \int \pi_{\text{naive}}(X_0, X_1)(X_1 - X_0)^2 dX_0 dX_1.\end{aligned}$$

So, basically you need to prove that the transport plan $\pi_{1\text{-rectified}}$ is better than π_{naive} .

Hint: Take a look at Theorem 3.5 on page 13 in <https://arxiv.org/pdf/2209.03003.pdf>. (1 point)

3. So, for the 1-rectified flow we started from just naive transport plan as our training data $(X_0, X_1) \sim \pi_{\text{naive}}$. Now instead of the naive transport plan $\pi_{\text{naive}}(X_0, X_1) = \pi_0(X_0) \times \pi_1(X_1)$ we will use the deterministic transport plan $\pi_{1\text{-rectified}}(X_0, X_1)$ defined by the deterministic transport map $T_{1\text{-rectified}}(Z_0)$ as

$$Z_1 = T_{1\text{-rectified}}(Z_0) = Z_0 + \int_{t=0}^1 v_\theta(Z_t, t) dt,$$

and the plan itself is

$$\pi_{1\text{-rectified}}(X_0, X_1) = \pi_0(X_0) \times \delta(X_1 - T_{1\text{-rectified}}(X_0))$$

to establish a better coupling (i. e. transport plan) between π_0 and π_1 than just naive coupling π_{naive} . We could then train the rectified flow to minimize the updated loss function

$$\min_{\theta} \int_0^1 \mathbb{E}_{(Z_0, Z_1) \sim \pi_{1\text{-rectified}}(Z_0, Z_1)} \left[\left\| Z_1 - Z_0 - v'_\theta(Z_t, t) \right\|^2 \right] dt,$$

with $Z_t = tZ_1 + (1 - t)Z_0$, to learn the 2-rectified flow $v'_\theta(Z_t, t)$. For the 3-rectified flow, the deterministic transport plan defined by v'_θ is used for further training, i. e. the corresponding transport map $T_{2\text{-rectified}}$ becomes

$$T_{2\text{-rectified}}(Z_0) = Z_0 + \int_{t=0}^1 v'_\theta(Z_t, t) dt,$$

and the transport plan becomes

$$\pi_{2\text{-rectified}}(X_0, X_1) = \pi_0(X_0) \times \delta(X_1 - T_{2\text{-rectified}}(X_0))$$

the loss function for the 3-rectified flow network $v''_\theta(Z_t, t)$ is

$$\min_{\theta} \int_0^1 \mathbb{E}_{(Z_0, Z_1) \sim \pi_{2\text{-rectified}}(Z_0, Z_1)} \left[\left\| Z_1 - Z_0 - v''_\theta(Z_t, t) \right\|^2 \right] dt.$$

You need to modify the following Colab notebook https://colab.research.google.com/drive/1CyUP5xbA3pjH55HDW0A8vRgk2EEyEl_P?usp=sharing#scrollTo=XmFYIjHqHN9c to train 1-, 2-, 3- rectified flows. Draw samples from the target distribution π_1 , which you will obtain by applying the transport map $T_{i\text{-rectified}}(Z_0)$, $i = 1, 2, 3$ $Z_0 \sim \pi_0$, note that it includes Euler numerical ODE solver inside. How does the samples from the target distribution change if you use $T_{2\text{-rectified}}$, $T_{3\text{-rectified}}$ for sampling? (1 point)

4. The main problem with this approach is that we use true samples from the target distribution π_1 only once when training the 1-rectified flow. In practice, this leads to the situation where the rectified flow ODE trajectories do not end at the target distribution $\pi_1(x)$, especially for the higher-order flows. Try training the 2-rectified flow using the inverse trajectory starting from Z_1 . Do you see any improvement? **(1 points)**